

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Quantum Mechanics      **L-T-P:** 3-1-0      **Credits:** 4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

**Syllabus:**

Stern-Gerlach experiment and spin, Dirac bra-kets, Hilbert space of state vectors; unitary transformations, Schrodinger, Heisenberg and Dirac (or Interaction) pictures. (8)

Approximation methods: the WKB approximation, Time independent perturbation theory (non-degenerate and degenerate cases), Zeeman effect and Stark effect, Spin-orbit coupling and fine structure of Hydrogen atom, Variational method, Helium atom. (12)

Quantum theory of scattering – cross sections, partial wave analysis, phase shifts, optical theorem. Schrodinger's equation as an integral equation, Green's function, Lippman Schwinger equation, Born's approximation, Yukawa potential, Coulomb scattering, Collisions of like particles. (12)

Time dependent perturbation theory, Fermi's Golden rule, Transition probabilities. Identical Particles and Spin. (10)

**Reference Books:**

1. Modern Quantum Mechanics: J J Sakurai
2. Quantum Mechanics: L.D. Landau and E.M. Lifshitz
3. Quantum Mechanics: L. I. Schiff

5. **Lecture-wise break-up:**

| Sl. No. | Topic                                                                                                                                                                                                                                                                                        | No. of lectures |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.      | Stern-Gerlach experiment and spin, Dirac bra-kets, Hilbert space of state vectors; unitary transformations, Schrodinger, Heisenberg and Dirac (or Interaction) pictures.                                                                                                                     | 8               |
| 2.      | Approximation methods: the WKB approximation, Time independent perturbation theory (non-degenerate and degenerate cases), Zeeman effect and Stark effect, Spin-orbit coupling and fine structure of Hydrogen atom, Variational method, Helium atom.                                          | 12              |
| 3.      | Quantum theory of scattering – cross sections, partial wave analysis, phase shifts, optical theorem. Schrodinger's equation as an integral equation, Green's function, Lippman Schwinger equation, Born's approximation, Yukawa potential, Coulomb scattering, Collisions of like particles. | 12              |

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|------------------------------|------------------------------------------------------------------------------------------------------------------|----|
| 4.                           | Time dependent perturbation theory, Fermi's Golden rule, Transition probabilities. Identical Particles and Spin. | 10 |
| <b>Total number of hours</b> |                                                                                                                  | 42 |